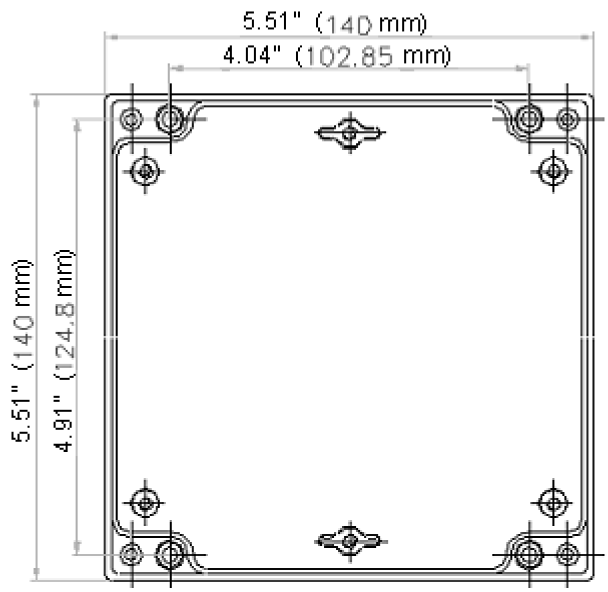
2.3 Mounting details for optional Oxygen Probe Interface unit, NX02INT

A Template for mounting is provided at the end of this manual for convenience. See Section 8.

The optional oxygen probe interface unit is designed to be fitted either within a control cabinet or without a control cabinet; the unit has a protection level of NEMA4 (IP65) providing suitable conduit glands are used. The interface unit can be mounted in any attitude, clearances should be maintained around the conduit entries to the unit to allow sufficient space for wiring etc. the ambient operating temperature range is 0 to 60°C (32 to 140°F). The unit MUST be grounded (earthed) to maintain electrical safety and ensure reliable operation.



Enclosure is 2.8" (71mm) deep

Power CANbus and
Probe cable entry

2.4 The oxygen trim option

> **CAUTION**
>
> - Use extreme care when handling the oxygen probe and wear heatproof gloves.
> - Ensure the burner is off before removing the oxygen probe from the flue.
> - If the boiler is to be operated with the probe removed, fit the blanking cover supplied since dangerous levels of carbon monoxide may be present in the flue.

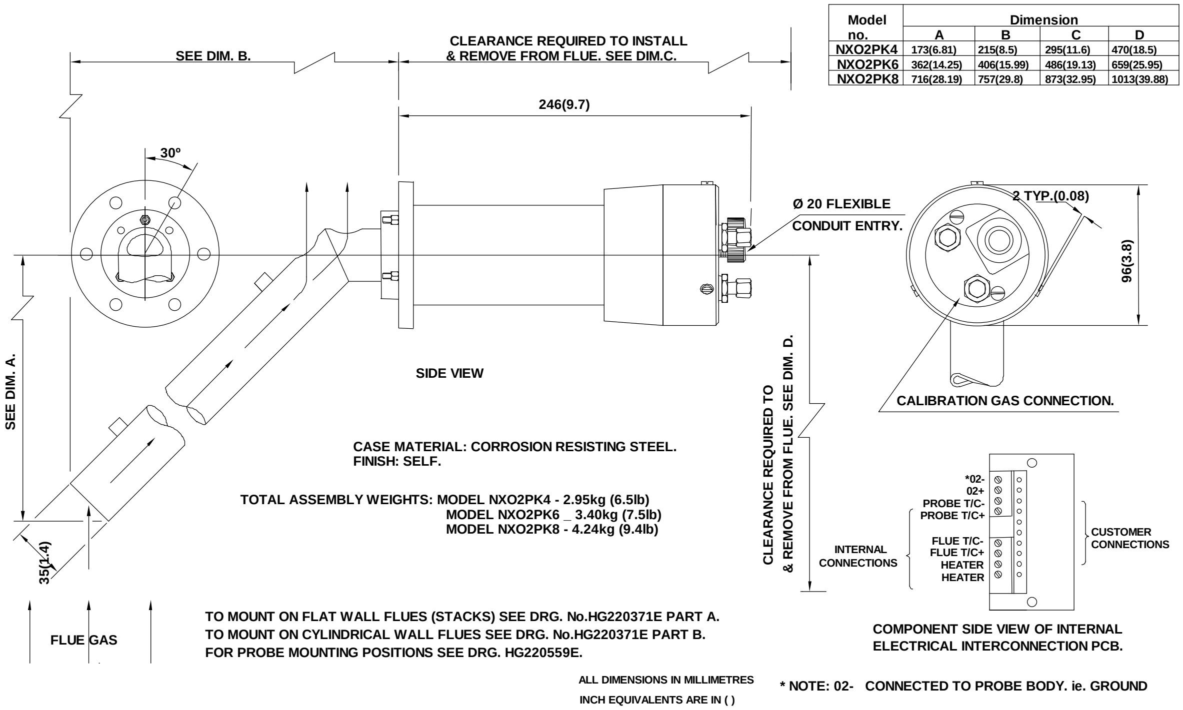
2.4.1 Oxygen Probe description

The oxygen trim / monitoring function is designed to be used with either an NXO2PK4, NXO2PK6, or NXO2PK8 oxygen probe. The probe offers fast, accurate response and good reliability when mounted in accordance with the guidelines in this section.

The probe is available in three different sizes.

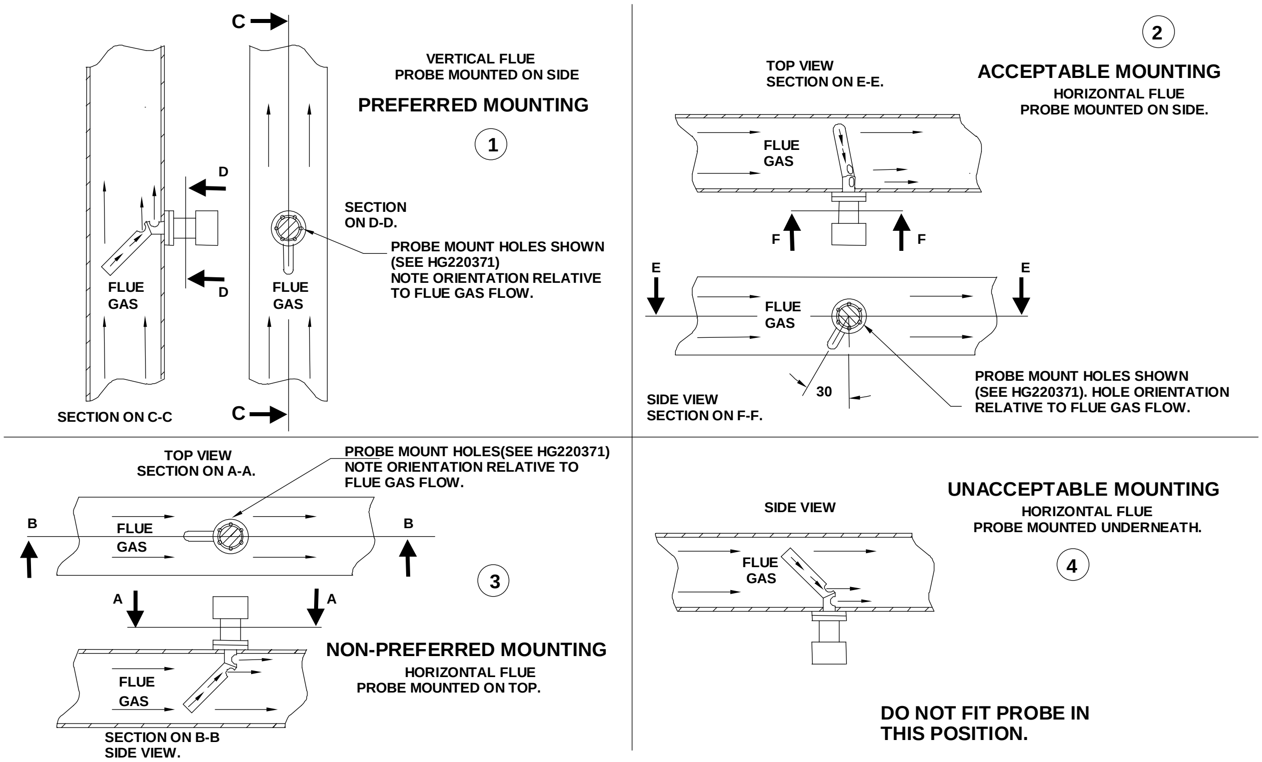
2.4.2 Installation of oxygen probe

Model no.	Dimension A	B	C	D
NXO2PK4	173(6.81)	215(8.5)	295(11.6)	470(18.5)
NXO2PK6	362(14.25)	406(15.99)	486(19.13)	659(25.95)
NXO2PK8	716(28.19)	757(29.8)	873(32.95)	1013(39.88)



2.4.3 Mounting the oxygen probe

The probe must be mounted in a manner that ensures that the flue gases pass into the gas tube at its open end and out of the tube at the flange end. Furthermore, if possible, the flange should be vertical with the gas tube angled downwards to ensure that particulates do not build up within the sample tube. Probe mounting with the flange horizontal is acceptable. Inverted probe mounting is not acceptable.

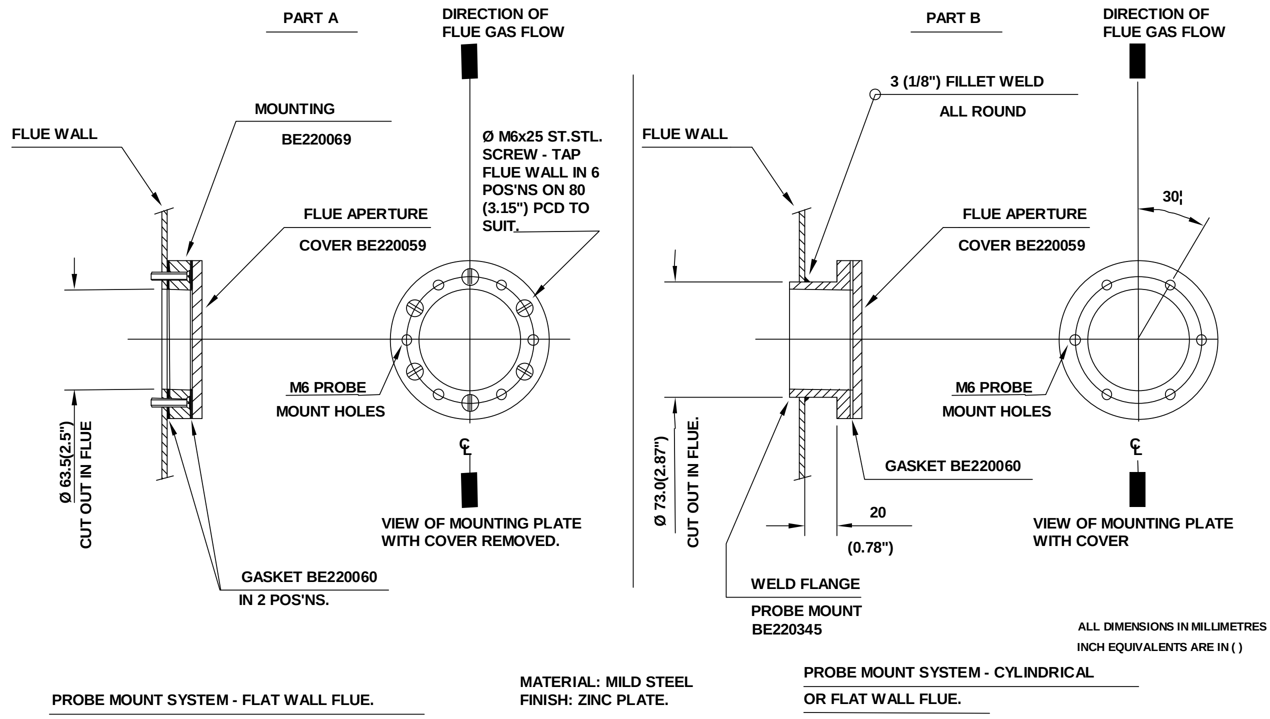


There are two types of flange available (see the drawing over the page). With either flange the vertical center line of the flange shown on the drawing should correspond to the gas flow direction.

6 stainless steel M6 x 20mm socket cap screws are provided for probe attachment.

The probe flange temperature must be maintained at the temperature of the flue wall by repacking or adding lagging, which may have been removed to mount the probe. Sulphate condensation will occur if the flue wall of an oil fired boiler falls below approximately 130°C. The sulphate problem does not occur in gas-fired installations, but vapor may cause problems due to condensation if the temperature of the flue gas falls below 100°C.

The maximum flue gas temperature is 1004ºF (540ºC).



`

The probe end cap carries a removable 20mm (3/4") flexible conduit fitting to enable probe replacement without wiring. The 2 hexagonal caps visible on the probe rear face are there to cover the calibration gas port and the sample gas port. The latter is merely a tube that passes directly into the flue to enable gas samples to be drawn or flue temperatures to be taken using other instrumentation. Both ports must be kept sealed during normal operation for safety and accurate performance.

2.4.4 Mounting arrangements for Temperature and Steam pressure sensors

For full technical specifications of sensors see sections 2.4.5 and 2.4.6

2.4.5 Boiler temperature sensors

½" Conduit Adaptor PN 35-336

The temperature sensor has a protection level of NEMA4, providing suitable conduit glands are used and can be mounted in any attitude. It has been designed for mounting into a well, or pocket, that has been inserted into the boiler shell. When choosing the position of the well, care should be taken to ensure that the sensor operates within its environmental specifications, and that the position will allow measurements, and subsequent control actions, to be correlated to other devices e.g. auxiliary safety stats.
The ambient operating temperature range is 0 to 60°C (0 to 140°F).
The unit MUST be grounded (earthed) to maintain electrical safety and ensure reliable operation.

2.4.6 Steam pressure sensors

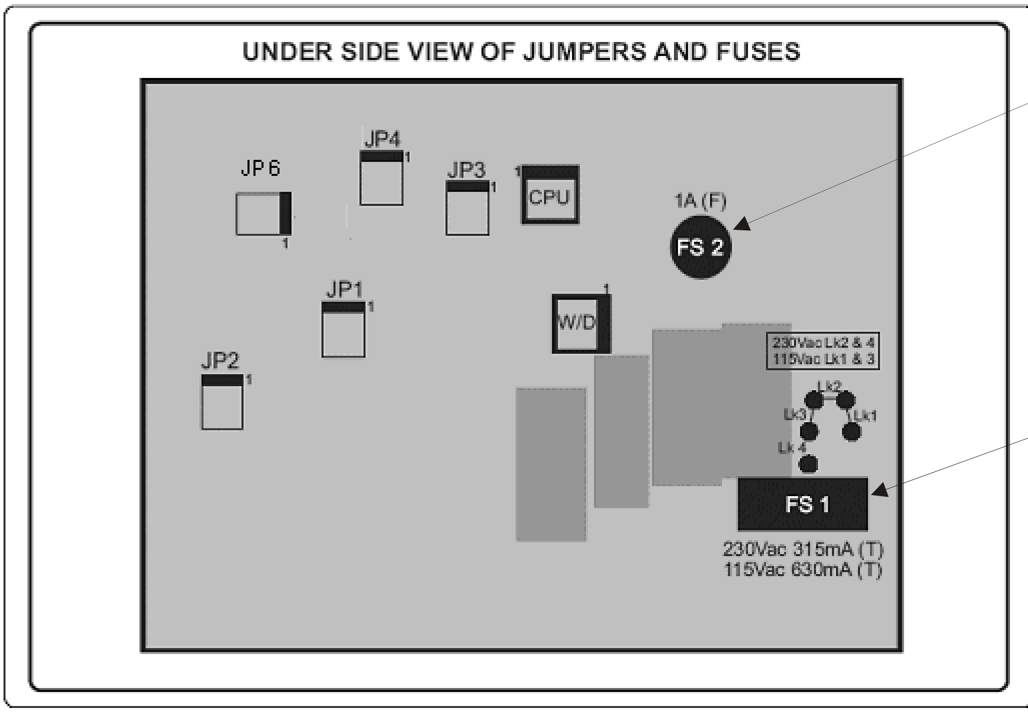
When fitting the sensor, care should be taken to ensure that the sensor operates within its environmental specifications. An important issue is the heating effect of the steam. Also, the sensor should be connected to the process in such a way that readings, and subsequent control actions, can be correlated to other devices e.g. the boiler pressure dial gauge and any auxiliary safety stats.

Steam Pressure sensors must be mounted in a vertical attitude to ensure water vapor does not collect inside the sensor. Additional devices, e.g. a “pig tail” feed pipe, may be required to reduce the possibility of moisture reaching the sensor during normal operation. Maintenance procedures should ensure that the sensor is inspected for evidence of condensates from the process collecting at the sensing point. If evidence of condensate is found, then preventative action must be taken to eliminate the cause.
The ambient operating temperature range is 0 to 70°C.
The unit MUST be earthed to maintain electrical safety and ensure reliable operation.

The conduit connection for steam pressure sensors is PG9, an adaptor for ½” NPSL is available.
Fireye ½” Conduit Adaptor P/N35-371.

2.5 Option link selection *(PPC6000)*

2.5.1 General (Access to jumper's and fuse's is gained by removing the back cover of the PPC6000)



Wickmann-Littlefuse
PN 3701100043
1 AMP (Fast Acting)
(to protect 5V DC circuit)
Alternate Source:
Mouser Electronics
PN 576-371100000

Bussmann - S504 series
P/N GMD-630 (Time Delay) for 115V
P/N GMD-315 (Time Delay) for 230V
(to protect 24V circuit)
Alternate Source:
Mouser Electronics
PN 504-GMD-630mA
PN 504-GMD-315mA

The PPC6000 has a number of option selection links, located on the circuit board. The function and settings are marked on the board alongside each link. **These links must be set to the correct position before power is applied to the control.** On some versions of the control additional details reference the option links and fuses are provided on a label attached to the mounting 'base', from which the product must be removed to gain access.

2.5.2 Line supply voltage (LK1 - 4) *(PPC6000)*

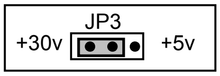
> ***WARNING***
>
> Incorrect setting of the Links **WILL** damage or destroy the unit.

The possible supply voltages are shown below, together with the necessary fuse rating. The correct fuse (type and rating) must be fitted; failure to do so may result in damage to the control.

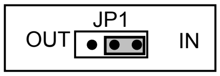
Supply voltage (V)	Links required	Fuse rating (mA)
120	LK1 and LK3	630 anti-surge – TIME DELAY
230	LK2 and LK4	315 anti-surge – TIME DELAY

2.5.3 SENS IN and SENS SUPP (boiler temp/pressure sensor) *(PPC6000)*

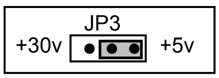
If a boiler pressure/temperature sensor or modulation potentiometer is used, links JP3 and JP1 must be set to suit the type of sensor and voltage requirement. For example:



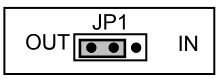
- For a 4-20mA loop-power sensor, choose a +30V supply.



- For a 4-20mA loop-power sensor, choose current (IN) input.



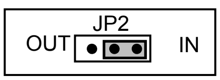
- For a 0-5V modulation signal, choose a +5V supply



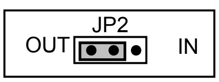
- For a 0-5V modulation signal, choose voltage (OUT) input

2.5.4 REMOTE SETPOINT *(PPC6000)*

If the remote setpoint or track signal is being used, link JP2 must be set to suit the type of signal. For example:

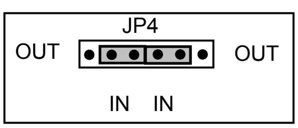


- For a current input set JP2 to IN position to ensure burden resistor *is* connected.



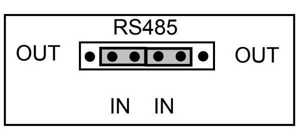
- For a voltage input signal set JP2 to OUT to ensure burden resistor *is not* connected.

2.5.5 RS485 serial communications termination resistor *(PPC6000)*

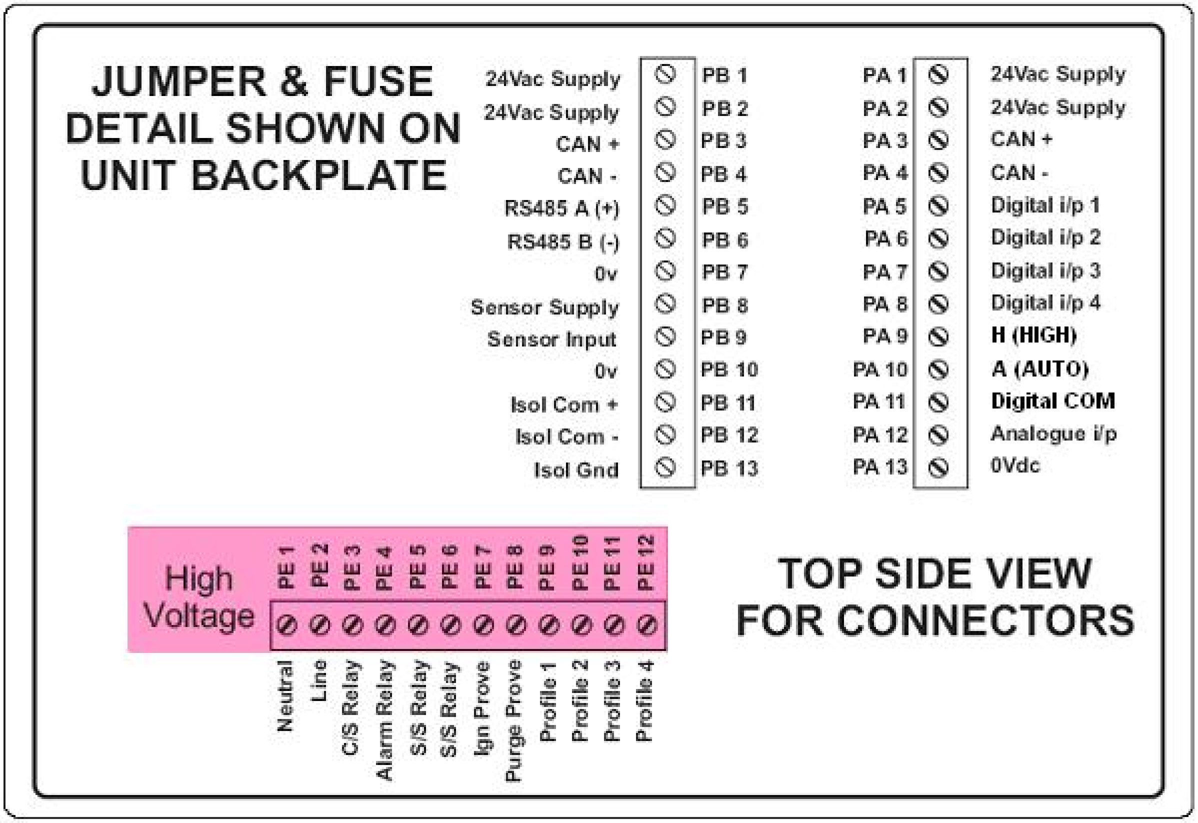


Please note that the RS485 serial communications supplied as part of the basic control is non-isolated, the termination resistor is selected by JP4. The two controls at the end of the communications bus should have this link set to the IN position. All other controls should have the link set to the OUT position. If only two controls are on the communications bus, set the links on both controls to the IN position.

2.5.6 RS485 serial communications termination resistor *(daughter board)*



The optional daughter board provides an isolated RS485 serial communication function, the termination resistor is selected by **the RS485 (JP2) jumpers on the daughter board, but the terminals are provided as part of the PPC6000.** The two controls at the end of the communications bus should have both of the RS485 (JP2) links set to the IN position. All other controls should have the links set to the OUT position. If only two controls are on the communications bus, set the links on both controls to the IN position.



Note: All wiring to terminals PA & PB are low voltage and must be braided shielded wire per table 2.6.1A. Wiring to terminals “PE” are line voltage. The maximum wire size is 16AWG (19.3mm) for all terminals.

Note: PE7 & PE8 provide purge and low fire position signals to the flame safeguard control. These outputs **MUST NOT** have a load greater than 30mA (i.e. relays, lamp, etc.), damage to the PPC6000 will result. This connection MUST be braided shielded wire.

2.6 Wiring

Typical Schematic

An abbreviated typical wiring schematic diagram showing the PPC6000 Parallel Positioning Controller, YB110 Flame Safeguard Control and YZ300 Expansion Module can be found in section 9.6in this manual.

This diagram is for reference only and may not meet all national or local codes. In all cases, local codes prevail with respect to the final installation of this product.

2.6.1 General

READ THIS FIRST!!!!

There are numerous mentions of "….overall braided shielded (screened) wire" throughout this manual. This is an important aspect to reliable operation. **Table 2.6.1-A** lists **the only approved wire** for this control. While one of the specifications relating to shielded wire indicates the amount of coverage (0-100%), this is not the only factor in selecting wire. While it is true, "foil and drain" shielded wire specifications indicate 100% coverage as compared to approximately 85% for braided type, the cross sectional area of the braid provides the required noise immunity. Also, the special grounding clamp bars on this control do not provide adequate connection to foil shield. In fact most foil shields do not conduct on the surface. **Using the "drain" wire to a ground stud does not properly protect the control.**

CAUTION

- Disconnect the power supply before beginning installation to prevent electrical shock, equipment and/or control damage. More than one power supply disconnect may be involved.
- Wiring must comply with all applicable codes, ordinances and regulations.
- Loads connected to the PPC6000 series, optional daughter board and optional oxygen probe interface must not exceed those listed in the specifications as given in this manual.
- Ensure the maximum total load on the CANbus cabling (servo-motors, display etc) is within the specifications of the PPC6000 and for the cable being used.
- This control MUST NOT be directly connected to any part of a Safety Extra Low Voltage (SELV) circuit.

WIRING INSTALLATION MUST BE CARRIED OUT BY A COMPETENT ELECTRICIAN AND IS SUBJECT TO I.E.E. WIRING REGULATIONS (BS 7671:1992), NEC AND/OR LOCAL STANDARDS, WHICH MAY PREVAIL.

HAZARDOUS VOLTAGES MUST BE ISOLATED BEFORE SERVICE WORK IS CARRIED OUT.

The PPC6000 unit MUST be mounted within a 'burner cabinet' or similar and MUST be grounded (earthed) to the overall enclosure to ensure safe and reliable operation.

Do not use a green or green/yellow conductor for any purpose other than ground (earth). The metal body of all component parts MUST be connected to ground (earth) using a green or green/yellow conductor.

The screen of the signal cable MUST not be used to provide the safety ground (earth), a separate connection using the largest cross-sectional area green or green/yellow ground (earth) wire possible MUST be made.

The screen termination clamps on the control are only provided to allow connection of the cable screens to the PPC6000 they do not provide strain relief. The signal cable screens MUST be connected at the screen termination clamps only, unless stated otherwise. **Screened cables MUST be of the 'copper braid shield' type** and not 'foil with drain wire', the cross section of the drain wire is insufficient to provide correct screening of the signals and there is also no provision to connect the foil or drain at the PPC6000.

Secure all cables carried in conduit at both ends using a suitable anchorage method in the cabinet.

All cabling that is required to operate at above 50v must be multi-strand single conductor (core), PVC insulated, 16 AWG (19/0.3mm) and should meet the requirements of I.E.C. 227 or I.E.C. 225, NEC

To comply with EMC requirements, wire the control and any optional units using the specified cable sizes and screen connections observing any maximum cable length limitations. **The manufacturer of this equipment recommends the use of bootlace ferules on all wire ends, as a "best practice".**

Bootlace
Ferules

The equipment described in this manual has been tested for compliance to the CE and UL directives listed in the section headed 'approvals'. However, once connected to a burner and other associated controls it is the responsibility of the installer to ensure the complete installation meets the requirements of the UL or CE directives relevant to the particular installation.

IMPORTANT: Wiring Guidelines

NOTE: Interposing terminal blocks should be avoided when shielded cable is required. Interposing terminals present a risk of electrical noise interference resulting in unreliable operation.

All wiring to this control *must* comply with National, State and Local electrical codes. In general, all insulation must meet or exceed the highest voltage present on any conductor in a conduit, raceway or panel, e.g. 480 volt motor wiring would require at least 600-volt insulation. Consult the National Electric code for guidance.

IMPORTANT: Low Voltage (vertical terminal strips)

All low voltage circuits and communication wire must be fully shielded braided type wire of the specified gauge and number of conductors. **Table 2.6.1-A provides the only approved wire for this application. No "or equal" is provided. Use of wire not approved by Fireye may VOID warranty.**

All wiring to terminal block "PA" & "PB" as well as to any optional daughter board (e.g. VSD) and the Power/CANbus wiring to the display, must be fully shielded braided wire per Table 2.6.1-A.

Under no circumstances should these input/outputs be connected to mains potential.
Connection of any voltage above 5 volts to these terminals **will** damage or destroy the unit.